

CyberLab IDS

Design and Implementation of a Virtual Network Intrusion Detection System Using pfSense and Suricata

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ABSTRACT

Cybersecurity has become one of the most significant challenges facing organizations in the digital era. Every day, networks are exposed to numerous attacks including reconnaissance scans, brute-force attacks, web application exploits, malware, and denial-of-service attacks. Detecting these attacks at an early stage is essential to minimize security risks and protect organizational assets.

This project presents the design and implementation of a complete virtual Intrusion Detection System (IDS) laboratory using pfSense Community Edition and Suricata IDS. The laboratory was implemented using Oracle VirtualBox to simulate a small enterprise network. pfSense was configured as the firewall and network gateway, while Ubuntu Linux acted as the attacking machine and Metasploitable2 served as the vulnerable target machine hosting the Damn Vulnerable Web Application (DVWA).

Several attack scenarios were performed to evaluate the effectiveness of the deployed IDS. Network reconnaissance was simulated using Nmap, while SQL Injection attacks were conducted against DVWA using SQLMap. Suricata continuously monitored network traffic and generated

security alerts that were analyzed to evaluate the detection capabilities of the IDS.

The project demonstrates how open-source security technologies can be combined to build an enterprise-grade security laboratory suitable for cybersecurity education, SOC analyst training, penetration testing awareness, and security monitoring research.

CHAPTER ONE

INTRODUCTION

1.1 Background

The rapid growth of information technology has significantly increased the dependence of organizations on computer networks and online services. As businesses continue to digitize their operations, the number and complexity of cyber threats continue to rise. Modern attackers employ sophisticated techniques to compromise systems, steal sensitive information, disrupt services, or gain unauthorized access to corporate networks.

Traditionally, firewalls have served as the first line of defense by filtering incoming and outgoing traffic based on predefined security policies. While firewalls are effective at controlling network access, they cannot always identify malicious activities hidden within legitimate network traffic.

To overcome this limitation, Intrusion Detection Systems (IDS) were introduced. An IDS continuously monitors network traffic, analyzes packets, compares them against thousands of attack signatures, and

generates alerts whenever suspicious behavior is detected. This allows security analysts to identify attacks before significant damage occurs.

Among the available IDS solutions, Suricata has become one of the most popular open-source security engines because of its high performance, deep packet inspection capabilities, protocol awareness, and multi-threaded architecture.

This project focuses on designing and implementing a complete IDS laboratory using pfSense Firewall integrated with Suricata IDS to demonstrate how organizations can monitor and detect malicious network activities within a virtual enterprise environment.

1.2 Problem Statement

Organizations continuously experience cyber threats targeting their internal infrastructure. Although firewalls provide essential access control mechanisms, they cannot detect every attack occurring inside allowed traffic. Modern attacks such as reconnaissance scans, SQL Injection, brute-force authentication attempts, malware communications, and application-layer attacks often bypass traditional firewall filtering.

Without an Intrusion Detection System, network administrators may remain unaware of malicious activities occurring inside their networks until after security incidents have already caused damage.

Therefore, implementing an IDS capable of monitoring network traffic and generating real-time alerts is essential for improving network visibility and strengthening organizational cybersecurity defenses.

1.3 Project Objectives

The primary objective of this project is to build a practical enterprise cybersecurity laboratory capable of detecting malicious network activities using open-source technologies.

The specific objectives include:

- Design a virtual enterprise network using Oracle VirtualBox.
- Deploy pfSense as the enterprise firewall.
- Configure WAN and LAN network interfaces.
- Configure DHCP, NAT, and DNS services.
- Install and configure the Suricata Intrusion Detection System.
- Update and manage Emerging Threats security rules.
- Simulate real cyberattacks within a controlled environment.
- Detect network reconnaissance attacks using Nmap.
- Analyze generated IDS alerts and log files.
- Document the entire implementation process for future reference.

1.4 Scope of the Project

This project is limited to designing and implementing a virtual enterprise security laboratory using open-source software. The laboratory focuses on network monitoring and attack detection rather than active attack prevention.

The implemented environment includes pfSense Firewall, Suricata IDS, Ubuntu Linux, Metasploitable2, and DVWA. Multiple network security scenarios are demonstrated, including reconnaissance detection and web application testing.

The project does not cover cloud-based deployments, advanced threat hunting, malware reverse engineering, or endpoint detection solutions.

1.5 Project Contributions

This project contributes to cybersecurity education by providing a complete practical implementation of a virtual IDS laboratory using freely available tools.

The major contributions include:

- Building a realistic enterprise network architecture.
 - Deploying an open-source firewall solution.
 - Integrating a high-performance IDS.
 - Simulating real-world cyberattack scenarios.
 - Demonstrating network monitoring and alert generation.
 - Producing comprehensive documentation suitable for educational and professional purposes.
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1.6 Importance of the Project

The importance of this project lies in providing hands-on experience with enterprise cybersecurity technologies commonly used by Security

Operations Centers (SOC). Rather than relying solely on theoretical knowledge, the project demonstrates the practical deployment, configuration, monitoring, and analysis of network security events.

The laboratory can serve as a foundation for future research involving Intrusion Prevention Systems (IPS), Security Information and Event Management (SIEM), threat intelligence integration, and advanced incident response.

CHAPTER TWO

Literature Review and Technology Overview

2.1 Introduction

Modern cybersecurity relies on multiple layers of defense rather than a single security solution. Firewalls, Intrusion Detection Systems (IDS), Intrusion Prevention Systems (IPS), endpoint protection, Security Information and Event Management (SIEM), and vulnerability assessment tools all work together to protect enterprise infrastructures.

This chapter introduces the core technologies used throughout this project and explains why each component was selected. Understanding the role of every technology provides the theoretical foundation required before discussing the practical implementation in later chapters.

2.2 Network Security

Network security is the process of protecting computer networks from unauthorized access, cyberattacks, data theft, service disruption, and malicious activities. As organizations increasingly depend on interconnected systems, securing network communication has become one of the highest priorities for IT departments.

Network security consists of multiple defensive mechanisms including:

- Firewalls
- Intrusion Detection Systems
- Intrusion Prevention Systems
- Virtual Private Networks (VPN)
- Network Segmentation
- Access Control
- Threat Monitoring
- Security Logging

Instead of relying on a single security device, organizations implement multiple layers of protection following the Defense in Depth strategy. This approach ensures that even if one security control fails, additional layers continue protecting the network.

2.3 Firewall

A firewall is considered the first line of defense in any network infrastructure. It examines incoming and outgoing network traffic and determines whether communication should be allowed or denied according to predefined security policies.

Modern firewalls perform several essential functions including:

- Packet filtering
- Stateful inspection
- Network Address Translation (NAT)
- VPN services
- Traffic routing
- DNS forwarding
- DHCP services
- Network segmentation

Although firewalls are extremely effective at restricting unauthorized network access, they cannot always detect attacks hidden inside allowed traffic. For example, if HTTP traffic is allowed through the firewall, malicious SQL Injection requests may also pass unnoticed. This limitation highlights the need for Intrusion Detection Systems.

2.4 Intrusion Detection System (IDS)

An Intrusion Detection System (IDS) is a security solution that continuously monitors network traffic in order to identify malicious or suspicious activities. Unlike traditional firewalls, an IDS does not primarily focus on allowing or blocking traffic; instead, it analyzes packets and generates alerts whenever suspicious behavior is detected.

IDS solutions generally rely on two detection techniques:

Signature-Based Detection

This method compares network traffic against a database of known attack signatures. If traffic matches an existing signature, an alert is generated.

Advantages:

- Fast detection
- Low false positive rate
- Excellent for known attacks

Disadvantages:

- Cannot detect unknown attacks
- Requires frequent signature updates

Anomaly-Based Detection

Instead of using predefined signatures, anomaly-based IDS systems establish a baseline of normal network behavior and identify deviations from that baseline.

Advantages:

- Can detect previously unknown attacks
- Detects insider threats

Disadvantages:

- Higher false positive rate
- Requires training period

Most enterprise IDS solutions combine both methods to improve detection accuracy.

2.5 Difference Between IDS and IPS

Although IDS and IPS share many similarities, they perform different roles inside enterprise networks.

Intrusion Detection System (IDS)		Intrusion Prevention System (IPS)
Monitors traffic		Monitors and blocks traffic
Generates alerts		Blocks malicious packets automatically
Passive security		Active security
No traffic modification		Can drop malicious packets
Suitable for monitoring		Suitable for prevention

In this project, Suricata was initially deployed in IDS mode to monitor network traffic and analyze alerts. This approach minimizes the risk of accidentally blocking legitimate traffic while learning network monitoring concepts.

2.6 Why pfSense?

pfSense Community Edition is one of the most widely adopted open-source firewall platforms available today. Built on the FreeBSD operating system, pfSense provides enterprise-grade firewall functionality without requiring expensive commercial licenses.

The following features make pfSense an excellent choice for cybersecurity laboratories:

- Stateful Packet Inspection Firewall
- NAT
- DHCP Server
- DNS Resolver
- VPN Support
- High Availability
- Package Manager
- Web-based Management Interface
- Integration with Suricata and Snort

Another significant advantage of pfSense is its intuitive web interface, allowing administrators to configure complex firewall policies without manually editing configuration files.

For these reasons, pfSense was selected as the firewall platform for this project.

2.7 Why Suricata?

Suricata is an open-source, high-performance Intrusion Detection and Intrusion Prevention System developed by the Open Information Security Foundation (OISF).

Unlike many legacy IDS solutions, Suricata supports multi-threaded packet processing, enabling it to inspect large volumes of network traffic efficiently on multi-core processors.

Suricata provides advanced capabilities including:

- Deep Packet Inspection (DPI)
- Protocol Identification
- HTTP Analysis
- TLS Inspection
- DNS Analysis
- SMB Inspection
- FTP Monitoring
- SMTP Inspection
- File Extraction
- JSON Logging (EVE JSON)
- Lua Scripting
- Multi-thread Processing

Suricata can operate in both IDS and IPS modes, making it suitable for enterprise security environments.

In this project, Suricata was configured to inspect LAN traffic and generate alerts whenever suspicious network activities were detected.

2.8 Emerging Threats Open Rules

Suricata relies on rule sets to identify malicious activities. One of the most popular free rule sources is Emerging Threats Open.

The Emerging Threats Open rule set contains thousands of signatures covering a wide range of attack categories, including:

- Malware Communication
- Port Scanning
- SQL Injection
- Web Attacks
- Botnet Activity
- DNS Attacks
- Exploitation Attempts
- Suspicious User Agents
- Command and Control Traffic
- Policy Violations

These rules are updated regularly by the cybersecurity community to ensure detection of newly discovered threats.

Throughout this project, the Emerging Threats Open rule set was installed and updated to provide comprehensive attack detection capabilities.

2.9 Oracle VirtualBox

Oracle VirtualBox is a virtualization platform that enables multiple operating systems to run simultaneously on a single physical computer.

Virtualization provides several advantages:

- Cost-effective laboratory environment
- Safe attack simulation
- Rapid deployment
- Snapshot support
- Easy recovery after testing
- Isolation from the host operating system

The complete cybersecurity laboratory developed in this project was implemented using Oracle VirtualBox.

2.10 Ubuntu Linux

Ubuntu Linux was selected as the attacker workstation because it provides a stable environment with extensive support for penetration testing and network analysis tools.

Ubuntu was used to execute:

- Nmap
- SQLMap
- Ping
- Web Browsing
- Network Testing

The operating system also provided Internet connectivity through pfSense, allowing realistic enterprise network simulations.

2.11 Metasploitable2

Metasploitable2 is an intentionally vulnerable Linux virtual machine designed for penetration testing practice and cybersecurity education.

The operating system contains numerous vulnerable services including:

- FTP
- SSH
- Telnet
- Apache
- MySQL
- Samba
- PostgreSQL
- Vulnerable Web Applications

Because of its intentionally insecure configuration, Metasploitable2 is widely used for ethical hacking laboratories.

Within this project, Metasploitable2 acted as the target machine.

2.12 DVWA (Damn Vulnerable Web Application)

DVWA is a PHP/MySQL web application intentionally designed with multiple security vulnerabilities. Its primary objective is to provide a legal environment for learning web application security.

DVWA includes several vulnerable modules such as:

- SQL Injection
- Command Injection
- Cross-Site Scripting (XSS)
- File Inclusion
- File Upload
- CSRF
- Brute Force Authentication
- Weak Session Management

During this project, DVWA was used to simulate SQL Injection attacks and generate web traffic for analysis by Suricata.

2.13 Nmap

Network Mapper (Nmap) is one of the most popular open-source network scanning tools available.

It is capable of:

- Host Discovery
- Port Scanning
- Operating System Detection
- Service Detection
- Version Detection
- NSE Script Execution

Within this project, Nmap was used to perform reconnaissance scans against the target machine. Suricata successfully detected these scanning activities and generated security alerts.

2.14 SQLMap

SQLMap is an open-source penetration testing tool specifically designed to automate SQL Injection testing.

Its capabilities include:

- Database Fingerprinting
- Data Extraction
- Authentication Bypass Testing
- Enumeration
- Database Dumping
- File System Access (when vulnerabilities exist)

SQLMap was used exclusively in the controlled laboratory environment to demonstrate web application vulnerability testing against DVWA.

2.15 Chapter Summary

This chapter introduced the theoretical concepts and technologies used throughout the project. It explained the importance of network security, described the roles of firewalls and intrusion detection systems, and justified the selection of pfSense, Suricata, VirtualBox, Ubuntu, Metasploitable2, DVWA, Nmap, and SQLMap. These technologies

collectively formed the foundation of the virtual cybersecurity laboratory implemented in this project.

CHAPTER THREE

System Analysis and Network Design

3.1 Introduction

Before deploying any cybersecurity solution, it is essential to understand the network architecture and identify how data flows between devices. A well-designed network topology simplifies security monitoring, improves traffic visibility, and allows security administrators to accurately identify malicious activities.

This chapter describes the architecture of the virtual enterprise network implemented in this project. It explains the function of each virtual machine, the communication process between devices, and the reasoning behind the selected topology.

The objective of this network design is to simulate a realistic enterprise environment where all network traffic passes through a centralized firewall equipped with an Intrusion Detection System (IDS). By forcing every packet to traverse the firewall, Suricata can inspect traffic, compare it against threat signatures, and generate alerts whenever suspicious behavior is detected.

3.2 Laboratory Architecture

The laboratory consists of three virtual machines running on Oracle VirtualBox. Each virtual machine performs a dedicated role similar to what would be found in a real enterprise network.

pfSense Firewall

The pfSense virtual machine acts as the network gateway and firewall. It connects the internal laboratory network to the external Internet through the WAN interface while simultaneously protecting the internal LAN network.

In addition to firewall functionality, pfSense hosts the Suricata Intrusion Detection System, allowing all packets entering or leaving the LAN to be inspected.

Primary responsibilities include:

- Packet Filtering
- NAT (Network Address Translation)
- DHCP Service
- DNS Resolution
- Routing
- Traffic Inspection
- Security Logging

Ubuntu Linux

Ubuntu represents an internal workstation.

During this project it was used to simulate user activities as well as attacker behavior.

Ubuntu was responsible for:

- Running Nmap
- Executing SQLMap
- Accessing DVWA
- Testing Internet Connectivity
- Verifying Firewall Rules

Using Ubuntu provides a safe and flexible environment for generating different types of network traffic while maintaining complete control over the laboratory.

Metasploitable2

Metasploitable2 represents a vulnerable server inside the enterprise network.

Instead of attacking real systems, all security testing was performed against this intentionally vulnerable operating system.

The server hosts several vulnerable services including:

- Apache Web Server

- MySQL Database
- FTP
- SSH
- Samba
- DVWA Web Application

The vulnerable web application was primarily used for SQL Injection testing.

3.3 Network Topology

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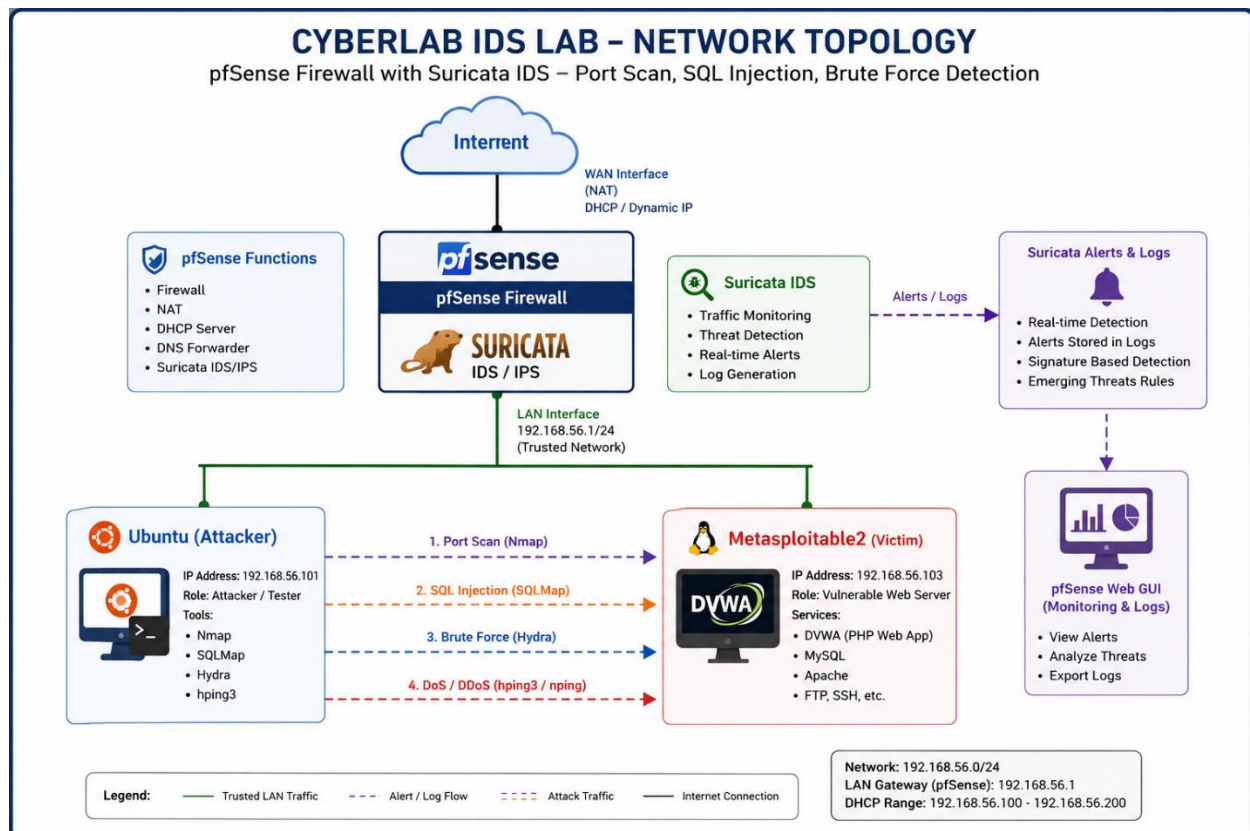


Figure 3.1 – Virtual Enterprise Network Topology

The implemented network topology follows a hub-and-spoke design where pfSense acts as the central gateway between the internal network and external connectivity.

All communications between Ubuntu and Metasploitable2 pass through pfSense, ensuring that Suricata is able to inspect every packet before it reaches its destination.

This architecture closely resembles enterprise deployments where firewalls are positioned at network boundaries.

3.4 Network Addressing

The laboratory used a private IPv4 addressing scheme.

Device	Interface	IP Address	Purpose
pfSense	WAN	DHCP	Internet Access
pfSense	LAN	192.168.56.1	Gateway
Ubuntu	LAN	192.168.56.101*	Client / Attacker
Metasploitable2	LAN	192.168.56.103*	Victim

*The Ubuntu and Metasploitable2 addresses may vary depending on DHCP assignment. Replace these values with the actual IP addresses from your environment if they differ.

3.5 Packet Flow

Understanding packet flow is essential when analyzing IDS alerts.

The communication process in the implemented laboratory follows these steps:

1. Ubuntu generates a network request.
2. The packet is forwarded to pfSense through the LAN interface.
3. Before forwarding the packet, Suricata analyzes it.
4. If the packet matches a security rule, Suricata generates an alert.
5. pfSense forwards the packet toward Metasploitable2.
6. The destination server processes the request.
7. The response travels back through pfSense.
8. Suricata inspects the response before it reaches Ubuntu.

Because every packet traverses the firewall twice, both inbound and outbound communications can be monitored.

3.6 Traffic Monitoring Workflow

The monitoring workflow implemented during this project can be summarized as follows:





This workflow illustrates the complete detection lifecycle, beginning with packet generation and ending with security event analysis.

3.7 Security Zones

The network can be logically divided into two security zones.

External Zone

The WAN interface connects pfSense to the Internet.

This interface receives traffic originating outside the laboratory.

Internal Zone

The LAN interface contains all internal devices.

Only trusted laboratory machines are connected to this network.

Suricata monitors traffic entering and leaving this zone.

3.8 Why This Architecture?

Several reasons motivated the selection of this architecture.

First, placing Suricata directly on pfSense eliminates the need for additional monitoring hardware.

Second, forcing all traffic through a single gateway ensures complete visibility into network communications.

Third, the architecture closely resembles real enterprise deployments where perimeter firewalls inspect incoming and outgoing traffic.

Finally, virtualization significantly reduces hardware costs while providing a realistic environment for cybersecurity experimentation.

3.9 Advantages of the Implemented Design

The implemented laboratory provides numerous advantages.

These include:

- Low deployment cost.
- Easy recovery using VirtualBox snapshots.
- Safe attack simulation.
- Centralized traffic monitoring.
- Flexible firewall management.
- Enterprise-like architecture.
- Scalability for future projects.
- Isolation from the host operating system.

These characteristics make the laboratory suitable for cybersecurity education, penetration testing practice, SOC analyst training, and security research.

3.10 Chapter Summary

This chapter presented the overall design of the virtual enterprise laboratory. It described the role of each virtual machine, explained the network topology, documented the IP addressing scheme, and analyzed the packet flow between network components. The implemented architecture provides a realistic platform for evaluating firewall policies and intrusion detection capabilities while maintaining a secure and isolated testing environment.

CHAPTER FOUR

System Implementation

4.1 Introduction

After completing the planning and design phases, the next step was implementing the virtual cybersecurity laboratory. This chapter describes the practical deployment process, including the installation of pfSense, network configuration, firewall setup, Suricata installation, IDS configuration, and laboratory verification.

The implementation was performed using Oracle VirtualBox on a Windows host operating system. Each component was configured individually before integrating the complete security environment.

The primary objective of this phase was to create a realistic enterprise network capable of monitoring internal traffic and detecting malicious activities.

4.2 Virtual Machine Deployment

The laboratory consisted of three virtual machines created using Oracle VirtualBox.

Firewall Machine

Operating System:

pfSense Community Edition

Purpose:

- Firewall
 - Gateway
 - DHCP Server
 - DNS Resolver
 - Intrusion Detection System
-

Client Machine

Operating System:

Ubuntu Linux

Purpose:

- Normal User Simulation

- Attack Simulation
- Network Testing
- Running Security Tools

Target Machine

Operating System:

Metasploitable2

Purpose:

- Vulnerable Server
- DVWA Hosting
- Attack Target



Figure 4.1 Oracle VirtualBox Virtual Machines

4.3 Installing pfSense

The pfSense firewall was installed from the official ISO image.

After booting the virtual machine, the installation wizard was launched.

The installation process included:

- Accepting the default keyboard layout
- Selecting automatic partitioning
- Installing the operating system
- Rebooting the firewall

After the installation completed successfully, pfSense automatically detected the available network adapters.

The WAN interface was configured to receive an IP address automatically using DHCP, while the LAN interface was configured manually using a private IPv4 address.

4.4 Network Interface Configuration

Two network interfaces were configured.

WAN Interface

Purpose:

Internet Connectivity

Configuration:

DHCP

LAN Interface

Purpose:

Internal Enterprise Network

Configuration:

Static Address

Example:

192.168.56.1/24

The LAN interface became the default gateway for all internal devices.

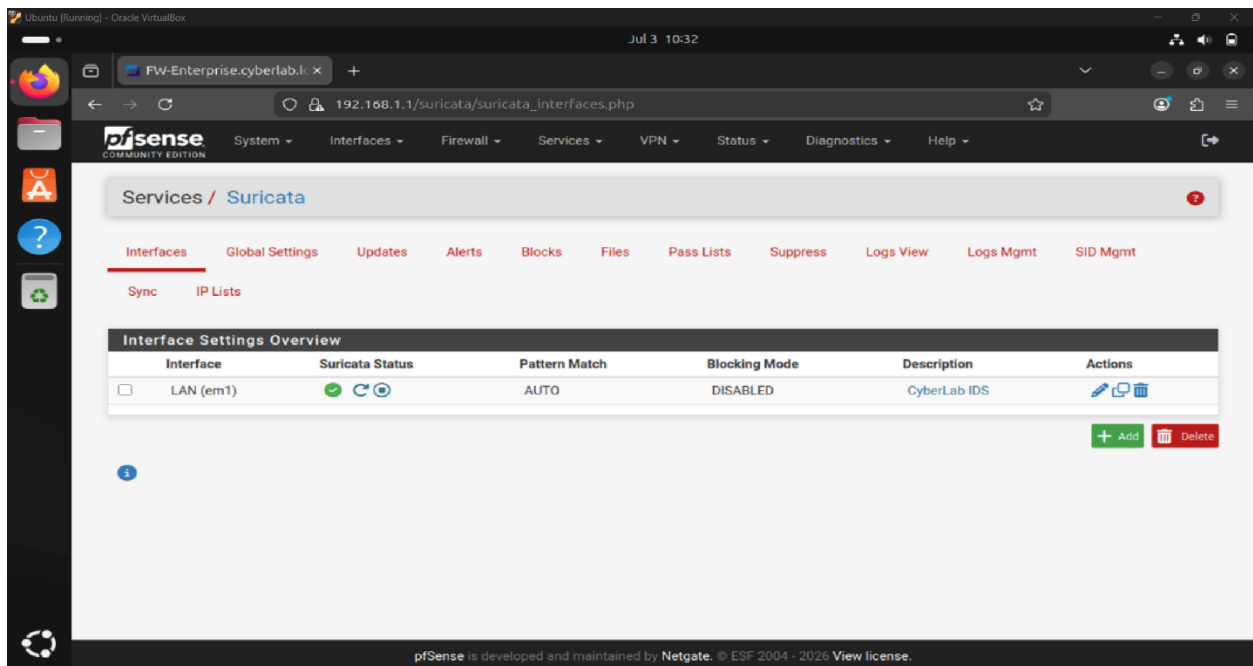


Figure 4.2 pfSense Interface Configuration

4.5 DHCP Configuration

To simplify network management, the DHCP Server was enabled on the LAN interface.

The DHCP server automatically assigned IP addresses to internal devices including Ubuntu and Metasploitable2.

This eliminated the need for manual network configuration.

The DHCP configuration included:

- DHCP Enabled
- Private Address Range
- Default Gateway
- DNS Server

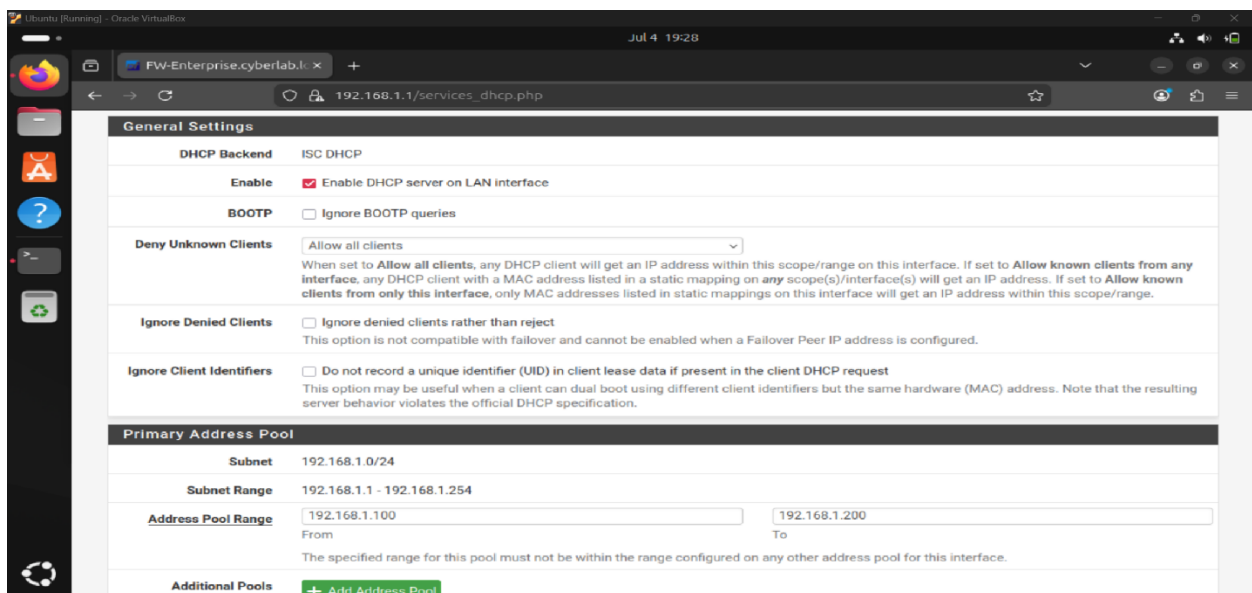


Figure 4.3 DHCP Configuration

4.6 DNS Configuration

Initially, DNS resolution failed because no valid DNS servers were configured.

As a result, the client machine was able to communicate using IP addresses but could not resolve domain names such as google.com.

The issue was solved by configuring reliable public DNS servers inside pfSense.

The following DNS servers were configured:

1.1.1.1

8.8.8.8

After applying the configuration, DNS resolution worked successfully and Internet access was restored.

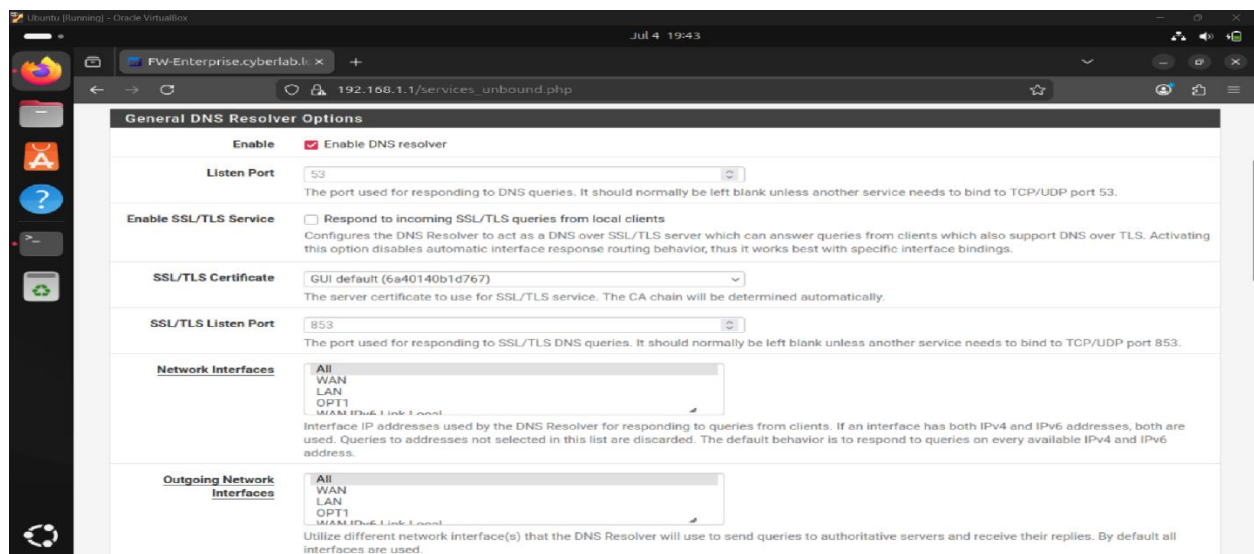


Figure 4.4 DNS Configuration

4.7 Firewall Rules

Firewall rules were configured to control traffic flowing through the LAN interface.

Instead of allowing unrestricted communication, custom rules were created for specific protocols.

The implemented rules included:

- ICMP
- HTTP
- HTTPS
- DNS

This approach demonstrates the principle of least privilege by allowing only the protocols required for laboratory operation.

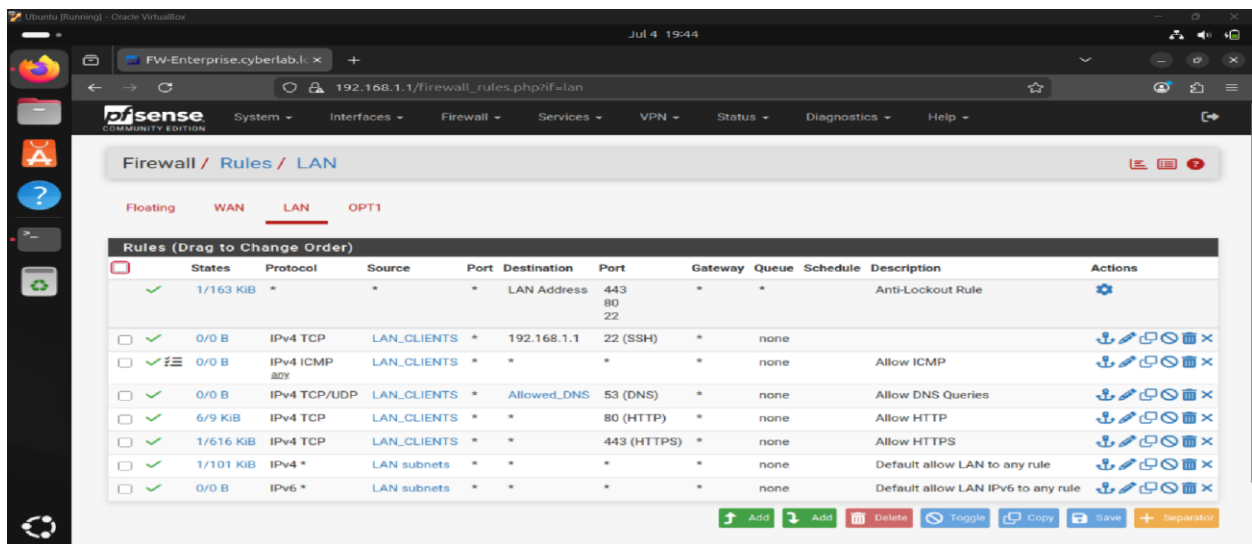


Figure 4.5 Firewall Rules

4.8 Installing Suricata

Suricata was installed using the pfSense Package Manager.

The installation process involved:

System



Package Manager



Available Packages



Searching for

Suricata



Installation

After installation completed successfully, Suricata became available inside the Services menu.

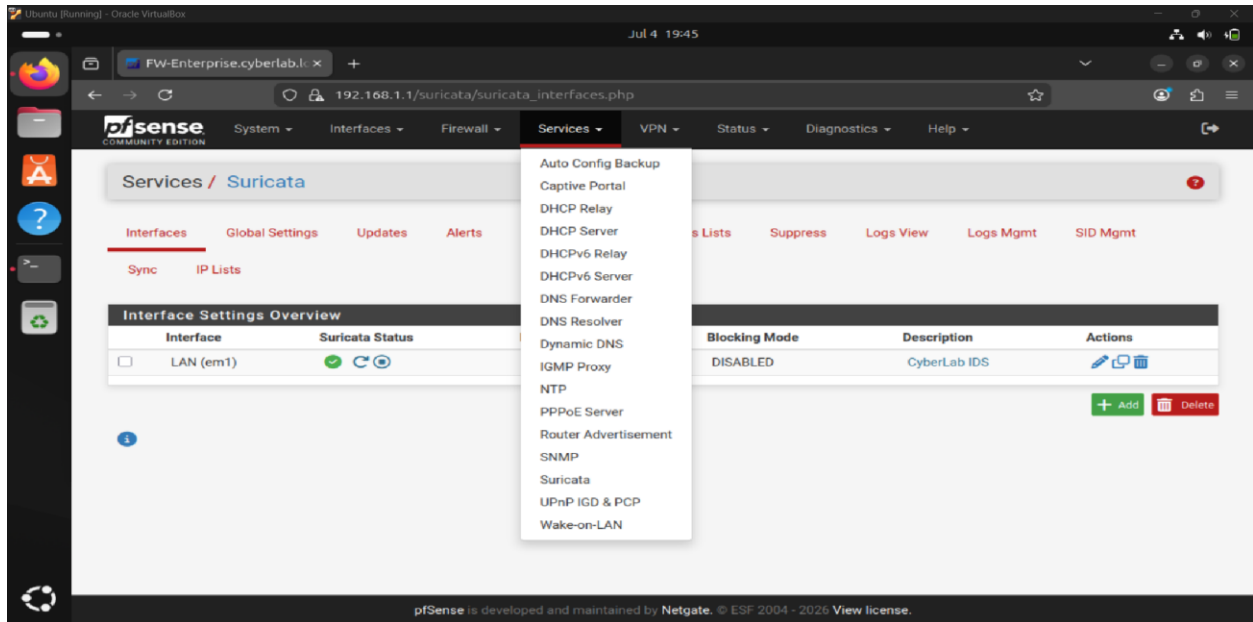


Figure 4.6 Suricata Installation

4.9 Updating Security Rules

An Intrusion Detection System is only effective when its rule database is continuously updated.

Therefore, the Emerging Threats Open rule set was enabled.

The update process downloaded thousands of attack signatures capable of identifying numerous network attacks.

The categories included:

- Network Scanning
- Web Attacks
- Malware
- Exploitation
- Policy Violations

- DNS Attacks
- Command and Control

Regular updates ensure that newly discovered attack signatures become available without reinstalling the IDS.

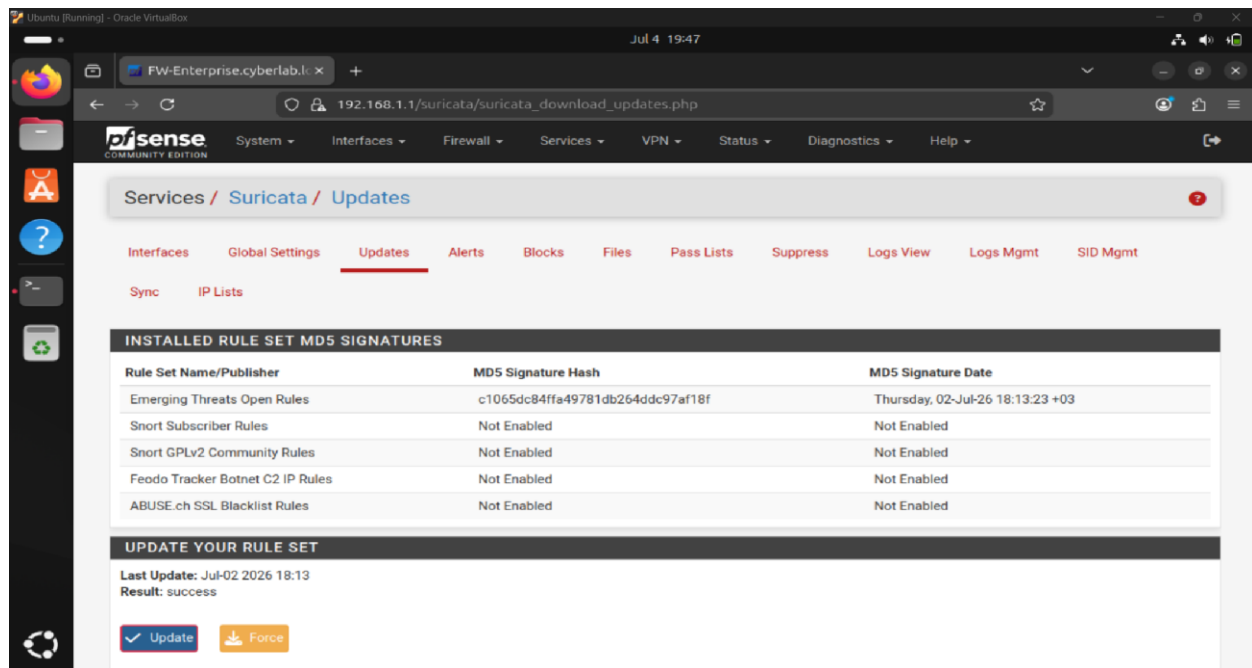


Figure 4.7 Suricata Rule Update

4.10 Configuring Suricata

Suricata was configured to inspect the LAN interface because all communications between Ubuntu and Metasploitable2 passed through this interface.

The following configuration was applied:

Interface: LAN

Inspection Mode: IDS

Rule Source: Emerging Threats Open

Logging: Enabled

Alert Generation: Enabled

The IDS continuously monitored all traffic traversing the LAN interface.

Whenever network packets matched a rule signature, an alert was generated automatically.

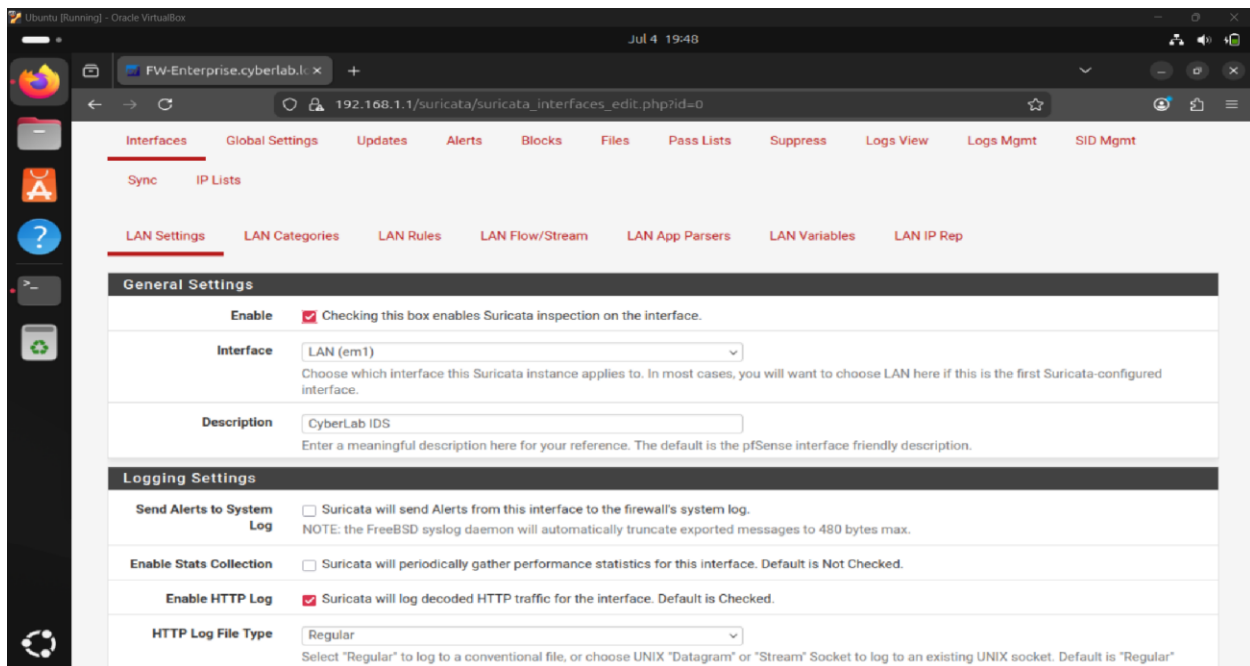


Figure 4.8 Suricata Interface Configuration

4.11 Verifying Connectivity

Before launching any attacks, network connectivity was verified.

Several tests were performed.

Test One

Ping between Ubuntu and pfSense

Result:

Successful

Test Two

Ping between Ubuntu and Metasploitable2

Result:

Successful

Test Three

Internet Connectivity

Result:

Successful

Test Four

DNS Resolution

Result:

Successful after configuring public DNS servers.

This verification ensured that all virtual machines could communicate correctly before beginning security testing.

4.12 Chapter Summary

This chapter documented the practical implementation of the virtual enterprise cybersecurity laboratory. It described the deployment of pfSense, network interface configuration, DHCP and DNS services, firewall policies, Suricata installation, and connectivity verification. Successfully completing these steps established a stable and functional environment suitable for conducting controlled cybersecurity experiments.

CHAPTER FIVE

Security Testing and Attack Simulation

5.1 Introduction

After completing the deployment of the virtual cybersecurity laboratory, several controlled attack scenarios were executed to evaluate the effectiveness of the implemented Intrusion Detection System (IDS). The objective of these tests was not to compromise the target system, but to verify whether Suricata could successfully identify suspicious activities occurring within the network.

All attack simulations were conducted inside an isolated virtual environment using intentionally vulnerable systems. No attacks were directed toward external systems or unauthorized targets. This approach ensured that all testing remained ethical, controlled, and compliant with cybersecurity best practices.

The attacks selected for this project represent common techniques frequently observed in enterprise environments. They include reconnaissance activities, web application testing, and network connectivity verification.

5.2 Testing Methodology

A structured methodology was adopted to ensure consistent and repeatable testing. Each attack scenario followed the same sequence of steps:

1. Prepare the laboratory and verify connectivity.
2. Launch the selected attack from the Ubuntu workstation.
3. Allow the traffic to pass through the pfSense firewall.
4. Inspect all packets using Suricata IDS.
5. Review generated alerts and log files.
6. Analyze the effectiveness of the detection rules.
7. Document the observations.

This methodology reflects the workflow commonly used by Security Operations Centers (SOC) when validating security controls.

5.3 Test Environment

The attack simulations were performed using the laboratory shown in Figure 5.1.

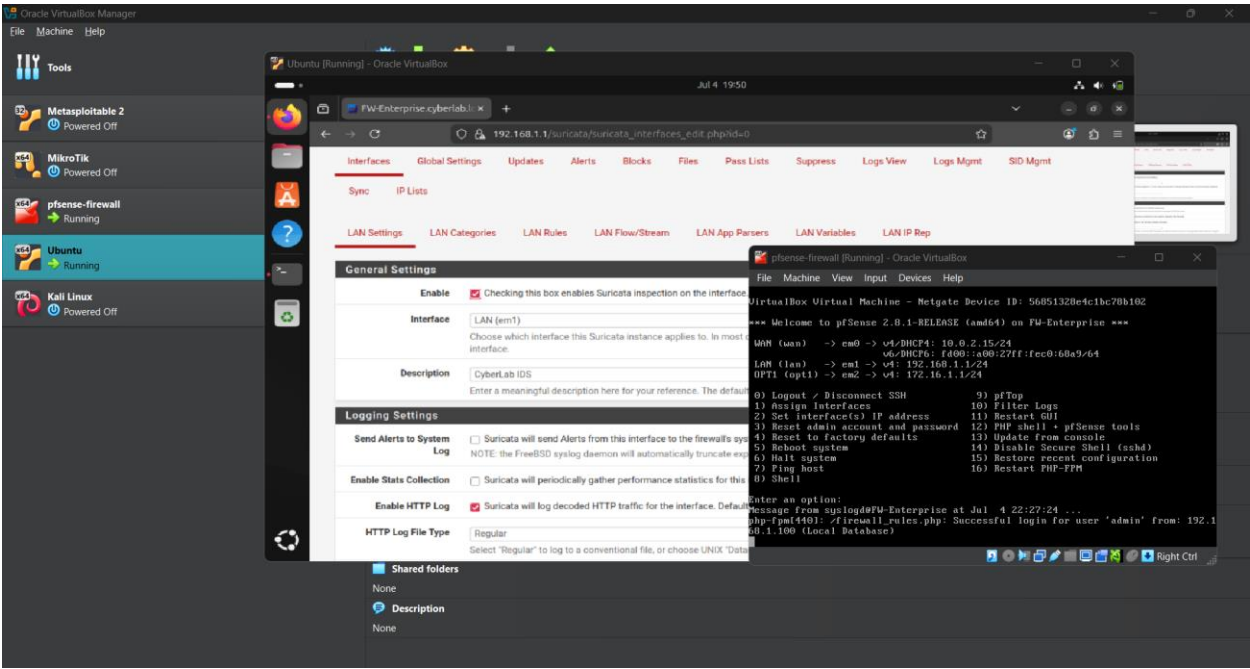


Figure 5.1 Laboratory Environment Before Security Testing

The environment included:

Device	Role
Ubuntu	Client / Attacker
pfSense	Firewall + IDS
Metasploitable2	Victim
DVWA	Vulnerable Web Application

5.4 Network Connectivity Verification

Before launching any attacks, basic network communication was verified.

The following tests were successfully completed:

- Ping to the pfSense gateway.
- Ping to the Metasploitable2 server.
- Internet connectivity verification.
- DNS resolution verification.

These tests confirmed that the laboratory environment was functioning correctly and that all traffic was traversing the firewall before reaching the destination.

5.5 Port Scanning Attack Using Nmap

One of the most common phases of any cyberattack is reconnaissance. Before exploiting a target, attackers typically identify active hosts, discover open ports, determine running services, and collect information about the operating system.

To simulate this phase, Nmap was executed from the Ubuntu workstation against the Metasploitable2 server.

The following command was used:

```
nmap -sS -A <Target_IP>
```

The SYN scan (-sS) performs a stealthy TCP scan to identify open ports, while the aggressive scan option (-A) enables service detection,

operating system fingerprinting, version detection, and default NSE scripts.

The scan identified several exposed services, including:

- FTP
- SSH
- HTTP
- MySQL
- Samba
- Telnet

These services are intentionally vulnerable in Metasploitable2 and provide a realistic target for laboratory testing.

5.6 IDS Detection of Port Scanning

While the Nmap scan was in progress, all traffic traversed the pfSense firewall where it was inspected by Suricata.

The IDS successfully detected the reconnaissance activity and generated multiple alerts associated with network scanning behavior. These alerts confirmed that the deployed rule set was capable of identifying suspicious scanning techniques commonly used during the reconnaissance phase of cyberattacks.

The generated alerts demonstrated that the IDS was functioning correctly and continuously monitoring traffic on the LAN interface.

5.7 SQL Injection Testing

Web application attacks represent one of the most common attack vectors targeting enterprise environments. SQL Injection occurs when user-supplied input is interpreted as part of a database query, allowing an attacker to manipulate the SQL statements executed by the server.

To simulate this attack, the Damn Vulnerable Web Application (DVWA) hosted on the Metasploitable2 server was used. The SQLMap tool was executed from the Ubuntu workstation to automate the testing process.

The attack targeted the SQL Injection module provided by DVWA in a controlled and isolated environment.

The testing process included:

- Accessing the vulnerable page.
- Establishing an authenticated session.
- Executing SQLMap.
- Sending crafted HTTP requests.
- Observing the responses.

5.8 Analysis of SQL Injection Results

SQLMap successfully generated HTTP requests containing SQL Injection payloads directed toward the DVWA application. The traffic reached the target server through the pfSense firewall and was inspected by Suricata.

During testing, not every SQL Injection request resulted in a dedicated IDS alert. This behavior is expected in practical environments because alert generation depends on several factors, including:

- The enabled rule categories.
- The specific Emerging Threats rule set version.
- The payload characteristics.
- The selected detection policies.

This observation highlights an important concept in intrusion detection: **the absence of an alert does not necessarily indicate the absence of malicious activity**. Security analysts should always correlate IDS events with application logs, firewall logs, and other security data sources before reaching conclusions.

5.9 Log Analysis

Suricata automatically recorded security events generated during testing.

The following log sources were examined:

- Alerts
- Fast Log
- EVE JSON
- Firewall Logs

Each log provides different information useful for security analysis.

The alert logs summarize rule matches, while EVE JSON contains structured event data suitable for integration with SIEM platforms such as Wazuh, Elastic Stack, or Splunk.

The collected logs demonstrated that Suricata continuously monitored network communications throughout the testing process.

5.10 Security Findings

The security assessment produced several important observations:

- The firewall successfully routed traffic while maintaining network isolation.
- Suricata continuously inspected all LAN traffic.
- Nmap reconnaissance activities generated IDS alerts.
- DNS configuration was critical for proper laboratory operation.
- Alert generation depended on the selected rule set and detection policies.
- Centralized logging simplified event analysis.

These findings demonstrate the effectiveness of integrating pfSense with Suricata for network monitoring.

5.11 Limitations

Although the laboratory successfully demonstrated intrusion detection capabilities, several limitations were identified:

- The environment consisted of a small number of virtual machines.

- SQL Injection detection depended on the available signatures.
- The IDS operated in passive detection mode rather than prevention mode.
- Limited system resources restricted the number of simultaneously running virtual machines.

These limitations do not reduce the educational value of the project but provide opportunities for future enhancement.

5.12 Chapter Summary

This chapter documented the practical security testing conducted within the virtual laboratory. Controlled attack simulations were executed using Nmap and SQLMap to evaluate the effectiveness of the deployed IDS. The generated alerts and collected log files confirmed that Suricata successfully monitored network activity and detected reconnaissance behavior. The chapter also highlighted the importance of interpreting IDS alerts alongside other security evidence to obtain a comprehensive understanding of network events.

CHAPTER SIX

Results, Discussion, Challenges, and Conclusion

6.1 Introduction

The primary objective of this project was to design, implement, and evaluate a virtual enterprise Intrusion Detection System (IDS) using pfSense and Suricata. Throughout the implementation process, several security tests were conducted to validate the effectiveness of the deployed environment.

This chapter presents the obtained results, discusses the effectiveness of the implemented solution, highlights the challenges encountered during development, and provides recommendations for future improvements.

6.2 Results

The virtual laboratory was successfully implemented and operated as intended. All core components functioned correctly after completing the network configuration and troubleshooting process.

The following objectives were successfully achieved:

Objective	Status
Virtual Network Deployment	Completed
pfSense Installation	Completed
WAN Configuration	Completed
LAN Configuration	Completed
DHCP Configuration	Completed
DNS Configuration	Completed

Objective	Status
Firewall Rules	Completed
Suricata Installation	Completed
Rule Updates	Completed
IDS Activation	Completed
Nmap Detection	Successful
Alert Generation	Successful
Log Collection	Successful

The results demonstrate that the implemented architecture met the primary objectives defined at the beginning of the project.

6.3 Discussion

One of the most significant observations made during this project is that implementing an IDS involves much more than simply installing software. A fully operational detection system requires proper network configuration, correct routing, reliable DNS services, appropriate firewall rules, and continuous validation.

During testing, Suricata successfully detected reconnaissance activities generated by Nmap. The generated alerts confirmed that the selected rule set was capable of identifying common network scanning techniques.

The SQL Injection experiment demonstrated another important lesson. Although SQLMap successfully generated attack traffic, IDS detection

depended on the enabled signatures and rule policies. This illustrates that no IDS provides perfect visibility under every circumstance. Security analysts must therefore combine IDS alerts with other evidence such as application logs, firewall logs, and endpoint telemetry.

Overall, the laboratory demonstrated that open-source tools can provide capabilities comparable to many commercial solutions when deployed and configured correctly.

6.4 Challenges Encountered

Throughout the implementation process, several technical challenges were encountered. Solving these issues significantly improved the understanding of enterprise networking and intrusion detection systems.

Challenge 1: DNS Resolution Failure

Initially, Ubuntu could communicate using IP addresses but failed to resolve domain names. As a result, Internet access appeared unavailable despite successful network connectivity.

Solution

The problem was resolved by configuring reliable public DNS servers within pfSense and verifying that clients obtained the correct DNS configuration through DHCP.

Challenge 2: Firewall Rule Configuration

Several firewall rules initially prevented expected communication between virtual machines. Incorrect protocol selection and missing DNS permissions resulted in connectivity problems.

Solution

Firewall rules were reviewed, simplified, and validated using connectivity tests until communication operated correctly.

Challenge 3: Suricata Configuration

Although Suricata was installed successfully, initial testing produced no alerts because traffic was not being inspected as expected.

Solution

The monitored interface was verified, the rule set was updated, and Suricata was restarted. After correcting the configuration, alerts were successfully generated.

Challenge 4: SQL Injection Detection

During SQL Injection testing, SQLMap successfully executed requests against DVWA, but corresponding IDS alerts were not always generated.

Solution

The rule configuration and available signatures were analyzed. It was concluded that alert generation depends on the enabled rules, the specific payloads used, and the selected detection policy. This reflects realistic operational environments where IDS visibility is influenced by configuration and rule coverage.

Challenge 5: Limited Hardware Resources

The project was implemented on a computer with limited RAM, restricting the number of virtual machines that could run simultaneously.

Solution

Virtual machines were started and stopped as needed, reducing memory consumption while allowing all experiments to be completed successfully.

6.5 Lessons Learned

This project provided valuable practical experience extending far beyond theoretical cybersecurity concepts.

The most important lessons learned include:

- Enterprise firewall configuration.
- Network troubleshooting techniques.
- DHCP and DNS management.
- NAT configuration.
- Intrusion Detection System deployment.
- Security rule management.
- Packet inspection principles.
- Log analysis.
- Alert interpretation.

- Attack simulation.
- Virtual laboratory construction.
- Documentation of cybersecurity projects.

The project also demonstrated the importance of patience and systematic troubleshooting when deploying complex security environments.

6.6 Future Improvements

Although the implemented laboratory successfully met its objectives, several enhancements could further improve its capabilities.

Future work may include:

1. Enable IPS Mode

Configure Suricata to operate in Intrusion Prevention System mode, allowing malicious packets to be blocked automatically rather than only generating alerts.

2. SIEM Integration

Forward Suricata logs to Security Information and Event Management platforms such as:

- Wazuh
- Elastic Stack (ELK)
- Splunk
- Graylog

This would enable centralized monitoring, correlation, and advanced visualization.

3. Additional Attack Scenarios

Expand the laboratory by testing:

- Brute Force Attacks
 - Cross-Site Scripting (XSS)
 - Remote Code Execution
 - DNS Tunneling
 - SMB Exploitation
 - Malware Simulation
 - Denial of Service (DoS)
-

4. Threat Intelligence Integration

Integrate external threat intelligence feeds to improve detection capabilities and identify known malicious IP addresses and domains.

5. Dashboard Development

Develop a real-time dashboard displaying:

- Alert statistics.
- Top attackers.
- Most targeted hosts.

- Attack timeline.
 - Protocol distribution.
 - Geographic visualization of external IP addresses.
-

6.7 Conclusion

This project successfully demonstrated the design and implementation of a virtual enterprise Intrusion Detection System using pfSense and Suricata.

The developed laboratory provided a realistic environment for studying enterprise network security while remaining completely isolated from production systems.

Throughout the implementation process, firewall configuration, network services, IDS deployment, and attack simulations were successfully completed. Reconnaissance traffic generated by Nmap was detected by Suricata, confirming the effectiveness of the deployed monitoring solution.

Beyond the technical implementation, this project strengthened practical skills in network troubleshooting, firewall management, intrusion detection, and security event analysis. The experience gained throughout the project closely resembles tasks commonly performed by Security Operations Center (SOC) analysts and network security engineers.

The project demonstrates that open-source technologies provide powerful capabilities for cybersecurity education, research, and professional skill development when combined with proper planning and implementation.

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Use the following references in APA style:

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APPENDIX A

Laboratory Screenshots

Insert the following screenshots:

- Figure A.1 VirtualBox Manager
- Figure A.2 Network Topology
- Figure A.3 pfSense Dashboard
- Figure A.4 Firewall Rules
- Figure A.5 DNS Configuration

- Figure A.6 Suricata Installation
 - Figure A.7 Rule Update
 - Figure A.8 Suricata Running
-

APPENDIX B

Commands Used During the Project

ping 192.168.56.1

ping 8.8.8.8

ping google.com

nmap -sS -A <Target_IP>

sqlmap -u

"http://<Target_IP>/dvwa/vulnerabilities/sqli/?id=1&Submit=Submit" --
batch

ip addr

ip route

cat /etc/resolv.conf

